

Class 11 Lab

Ian Koe

Background

PDB has 209,886 Entries (Oct/Nov 2025) UniPortKB has 199,579,901 entries

```
209886/199579901 * 100
```

```
[1] 0.1051639
```

PDB has only 0.1% coverage of the main sequence database

AlphaFold

3 main outputs

-predicted coordinates -local quality score -second quality score PAE

Run AlphaFold yourself if not happy with AFDB (no coverage, poor model)

Interpreting and Analyzing AF Results in R

```
results_dir <- "Dimer_23119.result/Dimer_23119"
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

# Print our PDB file names
basename(pdb_files)
```

```
[1] "Dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "Dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "Dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "Dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "Dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)

# Read all data from Models
# and superpose/fit coords
pdbbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

Reading PDB files:

```
Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
.....
```

Extracting sequences

```
pdb/seq: 1   name: Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_001_alphafold2_r
pdb/seq: 2   name: Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_002_alphafold2_r
pdb/seq: 3   name: Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_003_alphafold2_r
pdb/seq: 4   name: Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_004_alphafold2_r
pdb/seq: 5   name: Dimer_23119.result/Dimer_23119/Dimer_23119_unrelaxed_rank_005_alphafold2_r
```

```
rd <- rmsd(pdbbs, fit=T)
```

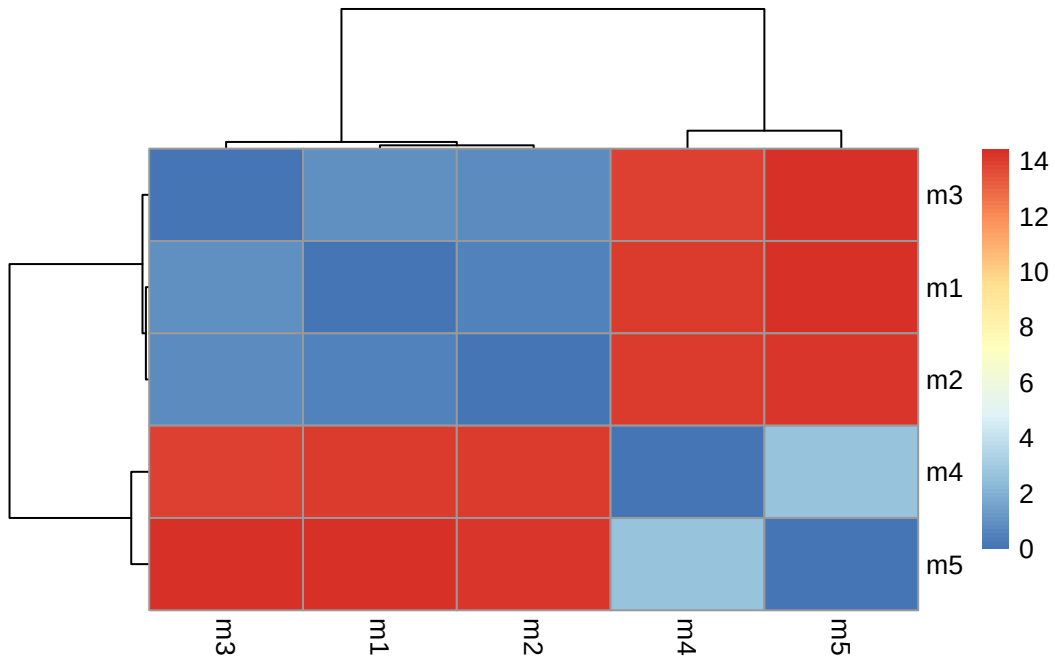
Warning in rmsd(pdbbs, fit = T): No indices provided, using the 198 non NA positions

```
range(rd)
```

```
[1] 0.000 14.428
```

```
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```



```
pdb <- read.pdb("1hsg")
```

Note: Accessing on-line PDB file

```
plotb3(pdb$b[1,], typ="l", lwd=2, sse=pdb)
points(pdb$b[2,], typ="l", col="red")
points(pdb$b[3,], typ="l", col="blue")
points(pdb$b[4,], typ="l", col="darkgreen")
points(pdb$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```

